MARINE eDNA METABARCODING WORKSHOP

Day1: NGS raw data → ASV

Marine eDNA Metabarcoding Analysis Using R

Jeju ARMS Case Study

Song, Chi-une

2026.08.31

WORKSHOP OVERVIEW

Overview — workshop workflow

From a marine sample to ecological interpretation

01 Sampling & DNA Preparation

Collect and prepare the biological material

1. Marine sample collection
2. DNA extraction
3. COI barcode amplification

02 NGS Sequencing

Convert amplified DNA into sequencing reads

1. Library preparation
2. Illumina sequencing
3. Raw sequencing output

03 Bioinformatics Processing

Turn raw reads into reliable sequence variants

1. FASTQ quality control
2. Filtering & denoising
3. ASV inference with DADA2

04 Ecological Analysis

Connect sequence variants to community ecology

1. Taxonomic assignment
2. Alpha & beta diversity
3. Ecological interpretation

DAY 1 · Raw reads → ASVs

DAY 2 · ASVs → Ecology

WORKSHOP OVERVIEW

Jeju ARMS Case Study

Do different marine habitats support different ARMS-associated communities?

GJ

Gangjeong

Macroalgae-dominated habitat

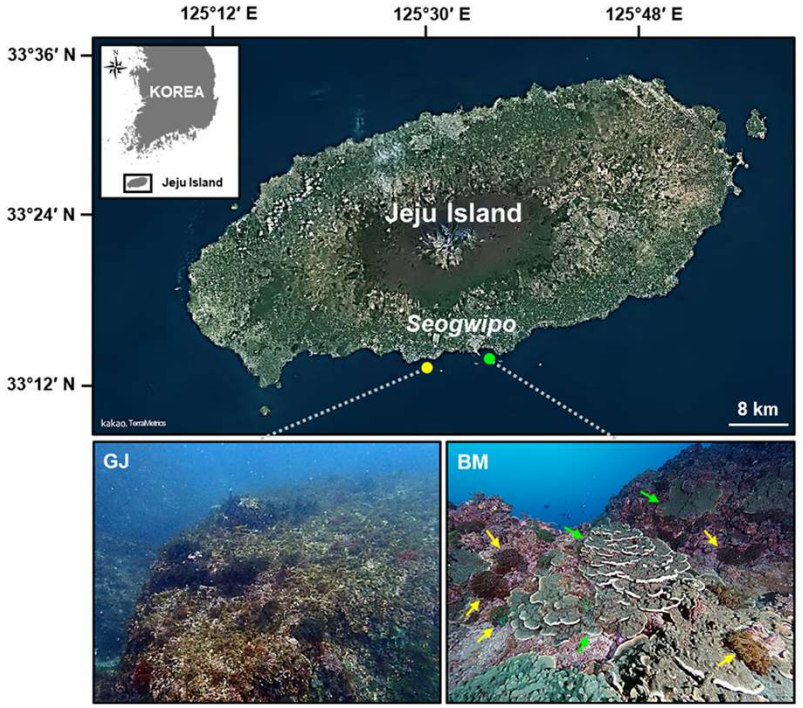
Turf-forming algae
Depth: ~13 m

BM

Bomok

Coral-dominated habitat

Scleractinian corals
Depth: ~12 m



Study sites on the southern coast of Jeju Island

Lee et al. (2024)

CORE QUESTION **Different surrounding habitats → Different ARMS communities?**

WORKSHOP OVERVIEW

Visual survey vs. DNA metabarcoding

Two complementary approaches

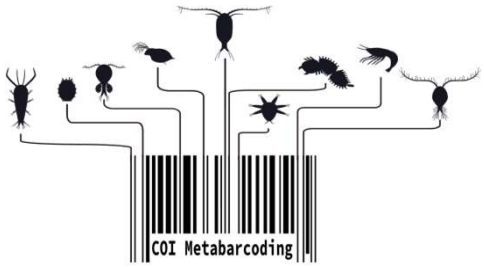
Visual survey

- Morphological identification
- Observer-dependent
- Limited resolution for some taxa

Lee et al. (2024)

DNA metabarcoding

- Higher sensitivity: Detect cryptic organisms
- More reproducible
- Potentially higher taxonomic resolution



DNA metabarcoding can detect biodiversity overlooked by visual surveys.

01

DNA barcode & NGS sequencing

- What is a DNA barcode?
- COI target region and primers
- PCR product to sequencing library
- Illumina paired-end sequencing

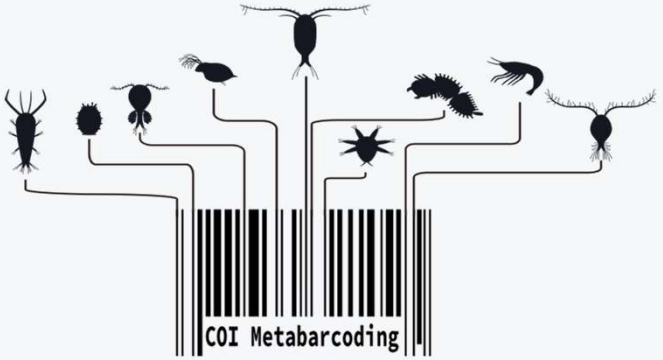
01

What is a DNA barcode?

Why do we target specific DNA regions?

Common barcode markers

- 16S rRNA — bacteria and archaea
- 18S rRNA — broad eukaryotic diversity
- COI — animals
- ITS — fungi



A useful DNA barcode should:

- be **amplifiable across many taxa**
- contain **enough sequence variation to distinguish taxa**
- have **well-represented reference database**

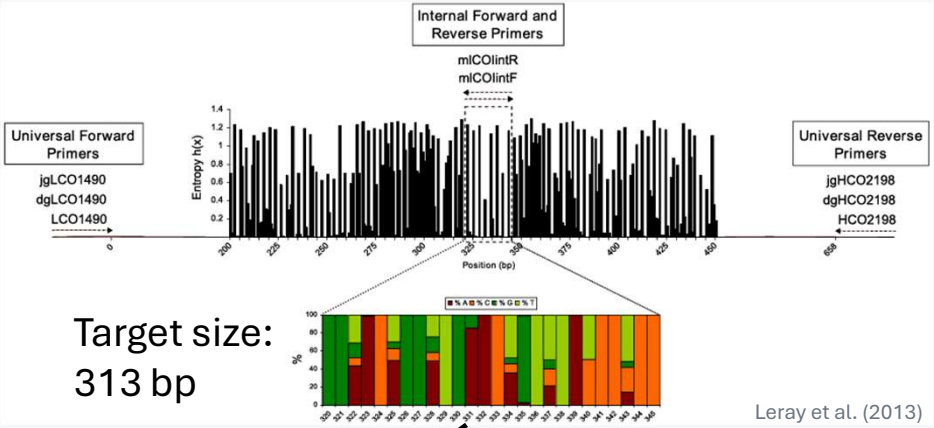
KEY POINT · A DNA barcode is a standardized genetic region used to distinguish and identify organisms.

COI target region and primers

Target marker: mitochondrial cytochrome c oxidase subunit I (COI)

Why COI?

- Widely used as the standard DNA barcode for animals
- Provides high taxonomic resolution for many animal groups
- Well represented in public reference databases



Leray et al. (2013)

COI barcode region

- Mitochondrial protein-coding gene
- Full COI gene: approximately **1.5 kb**
- Standard animal barcode region: approximately **658 bp**

Primer pair used in this workshop

- **mlCOIintF (Forward)**
  `5'-GGWACWGGWTGAACWGTWTAYCCYCC-3'`
- **jgHCO2198 (Reverse)**
  `5'-TAIACYTCIGGRTGICCRAARAAYCA-3'`

National Center for Biotechnology Information
NCBI

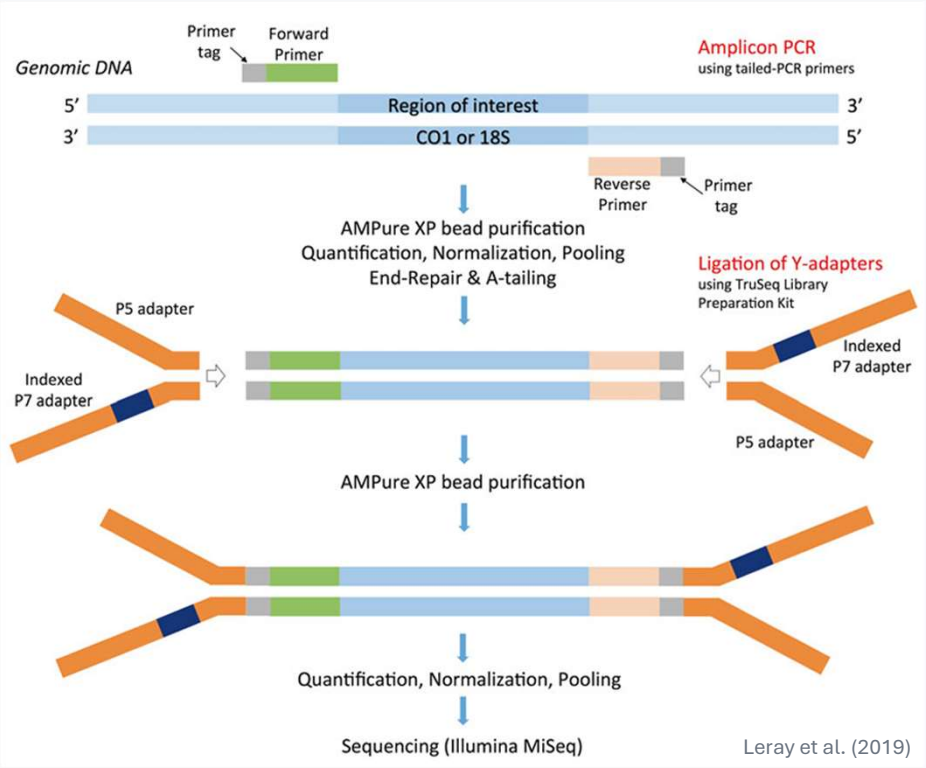
BOLDSYSTEMS

KEY POINT · COI combines sequence variation with extensive reference data, making it a widely used barcode for animals.

From PCR product to sequencing library

How do we prepare DNA fragments for Illumina sequencing?

- **PCR:** amplifies the target COI barcode region
- **Adapter:** allows DNA fragments to bind to the Illumina flow cell
- **Index:** identifies the sample of origin after sequencing



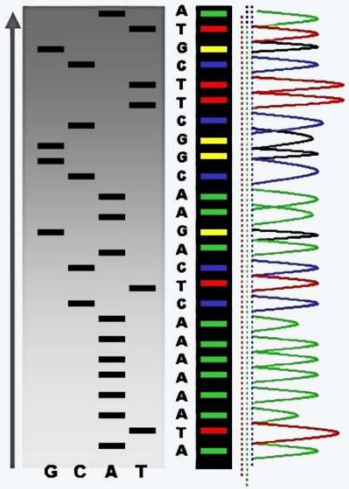
Leray et al. (2019)

KEY POINT · PCR selects the target; adapters enable sequencing; indexes identify samples.

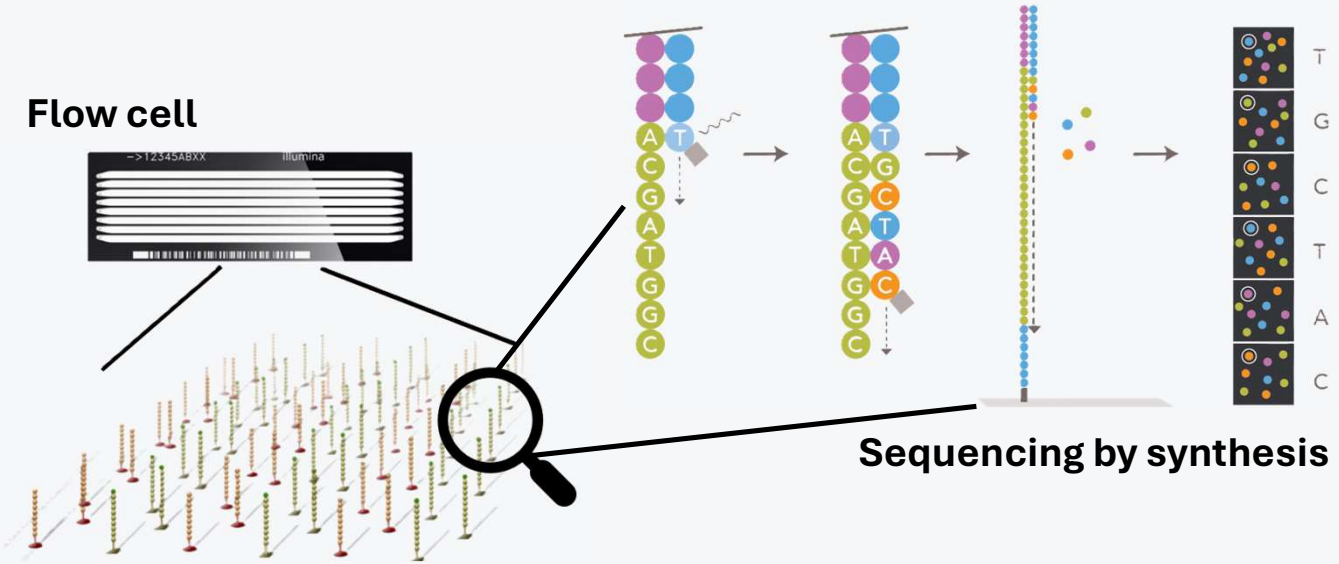
Illumina sequencing

How can millions of DNA fragments be sequenced at once?

- **Sanger sequencing:** sequences individual DNA templates separately
- **Illumina sequencing:** sequences millions of DNA fragments in parallel on a **flow cell**



Sanger sequencing



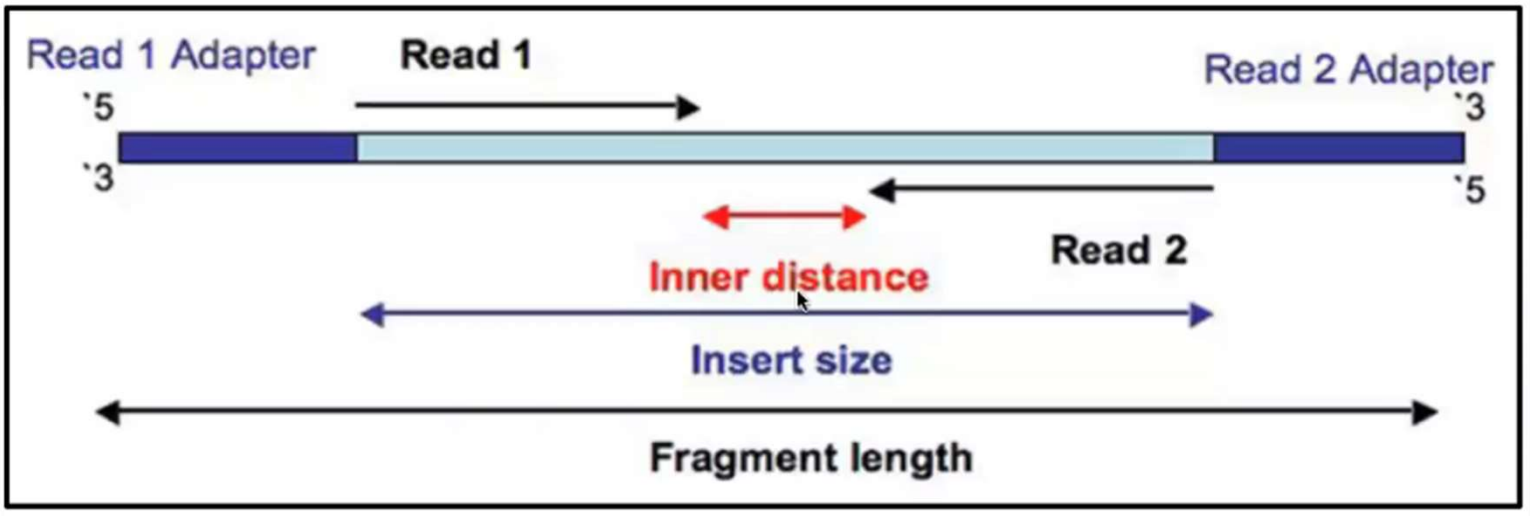
Illumina sequencing

KEY POINT · Illumina sequencing generates millions of reads in parallel.

Illumina sequencing

How does paired-end sequencing work?

- **Both ends of the same DNA fragment are sequenced.**
- **Read 1 (R1):** forward read
- **Read 2 (R2):** reverse read
- **Overlapping regions can be merged into a single sequence.**



Paired-end sequencing

KEY POINT · Paired-end sequencing reads the same DNA fragment from both ends, allowing overlapping reads to be merged.

02

BIOINFORMATICS PROCESSING

From raw sequencing reads to reliable ASVs

02

Raw sequencing output

What does the sequencing company actually give us?

Raw data statistics

Sample ID	Total read bases (bp)	Total reads	GC (%)	AT (%)	Q20 (%)	Q30 (%)
Bomok1	7,021,126	23,326	39.44	60.57	91.00	71.50
Bomok2	6,162,674	20,474	42.53	57.48	91.50	72.00
Bomok3	7,650,216	25,416	40.04	59.96	92.50	72.50
Bomok4	8,443,050	28,050	41.30	58.71	91.00	71.00
Gangjeong1	6,431,166	21,366	40.59	59.42	91.50	72.50
Gangjeong2	8,019,844	26,644	38.65	61.36	93.00	73.50
Gangjeong3	9,118,494	30,294	41.24	58.77	91.50	72.50
Gangjeong4	9,118,494	30,294	41.24	58.77	91.50	72.50

File contents

Download link	File size	md5sum
Bomok1_R1_001.fastq.gz	2.9 M	fb039b00b39bf46efe3b7db3bed9bf8b
Bomok1_R2_001.fastq.gz	3.2M	2cec4e1eede1a873a4fd24336a307033
Bomok2_R1_001.fastq.gz	2.6M	fad0e3e992afb749e4ffb3bd5dc11204
Bomok2_R2_001.fastq.gz	2.8M	62f0b05d654ce3907e1c5493aae669b6
Bomok3_R1_001.fastq.gz	3.2M	75615eb71d1309c74c8b2124b02a6c60
Bomok3_R2_001.fastq.gz	3.5M	5ffc12abf2dcb8bd7c4cf0a2e077a5ed
Bomok4_R1_001.fastq.gz	3.6M	9e4aeae7426ffee10d190da6c506266b
Bomok4_R2_001.fastq.gz	3.9M	80d7d19053393ea496abf3f0624e2f61
Gangjeong1_R1_001.fastq.gz	2.7M	f9f349e4f5a96202ecb49c88ad0cdf86
Gangjeong1_R2_001.fastq.gz	2.9M	73bcfe986778ec2a6043728ab7a624ed
…		

- **Sequencing output**

  Sample A

  → **R1:** Sample_A_1.fastq.gz

  → **R2:** Sample_A_2.fastq.gz

- **Sequencing statistics:**
  - read counts
  - total bases
  - `GC/AT` content
- **Quality metrics:** `Q20` and `Q30`

KEY POINT · Raw sequencing output includes paired FASTQ files and basic sequencing-quality information.

What is inside a FASTQ file?

How are sequence and quality information stored?

- Each sequencing read is stored as a four-line FASTQ record.

FASTQ format

```
@SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=72
GGGTGATGGCCGCTGCCGATGGCGTCAAATCCCACCAAGTTACCCTTAACAACTTAAGGGTTTTCAAATAGA
+SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=72
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!9!G9!C!!!!!!!!!!!!!!!!!!!D!!!!!!!>!!!!!!/
```

Field 1
Field 2
Field 3
Field 4

Quality scores are encoded as `ASCII` characters.

Line	Contents
1	@ Read identifier
2	DNA sequence
3	+ Separator
4	Base-quality scores corresponding to each nucleotide in Line 2

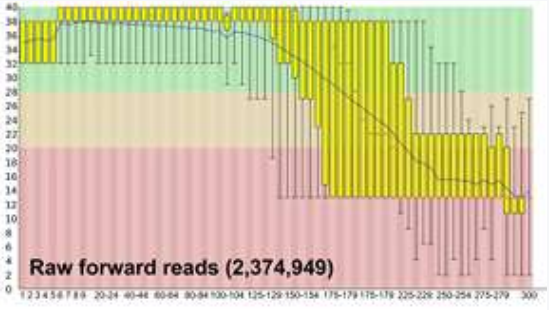
KEY POINT · Every FASTQ read contains both sequence and quality information.

What is a Phred quality score?

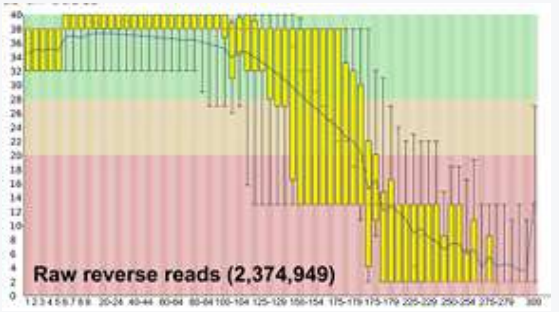
How confident are we in each base call?

- Higher Q = higher confidence in the base call
  - Q20: ~99% base-call accuracy
  - Q30: ~99.9% base-call accuracy
- Modern Illumina FASTQ files commonly use Phred+33 encoding
- Read quality often declines toward the end of a read

Read1 (forward)



Read2 (reverse)



Why does this matter?

- Low-quality bases can create false sequence variants.

KEY POINT · Higher Phred scores mean more reliable nucleotide calls.

03

Practice

R/R studio setup
FASTQ quality control
ASV inference

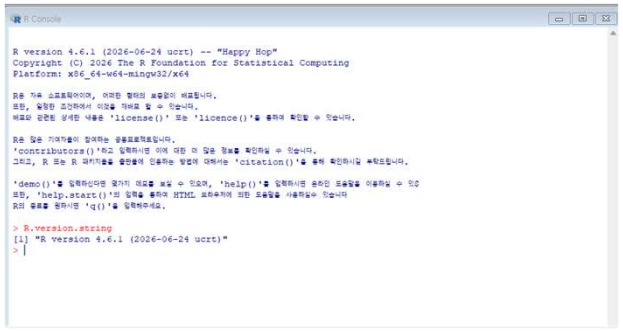
03

03 PRACTICE

R and RStudio

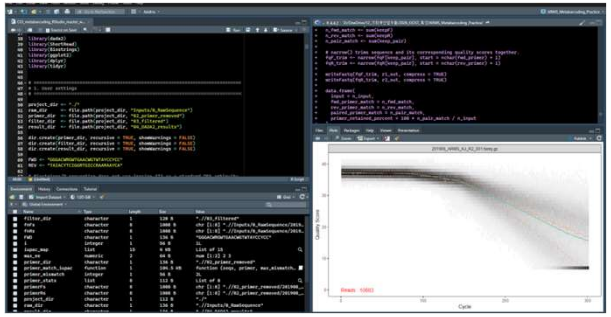
Why do we use them for metabarcoding analysis?

- **Free & open-source programming language and environment** for statistical computing
- Runs on **Windows, macOS,** and **Linux**
- Large ecosystem of packages for **bioinformatics, statistics,** and **ecology**
- Script-based analysis supports **reproducibility**



R Console

- An **integrated development environment (IDE)** for R
- Write and run R code directly from the source editor
- Organizes **scripts, files, plots, objects,** and **help** in one workspace
- **Projects** help organize analysis files and working directories



RStudio

KEY POINT · R performs the analysis; RStudio provides the workspace for working with R.

03 PRACTICE

Installing R on Windows

What do we need before using RStudio?

1 DOWNLOAD

Download R for Windows → base
Click "Download R-4.6.1 for Windows"

* Note: macOS users should download the ***macOS version***.

2 INSTALL

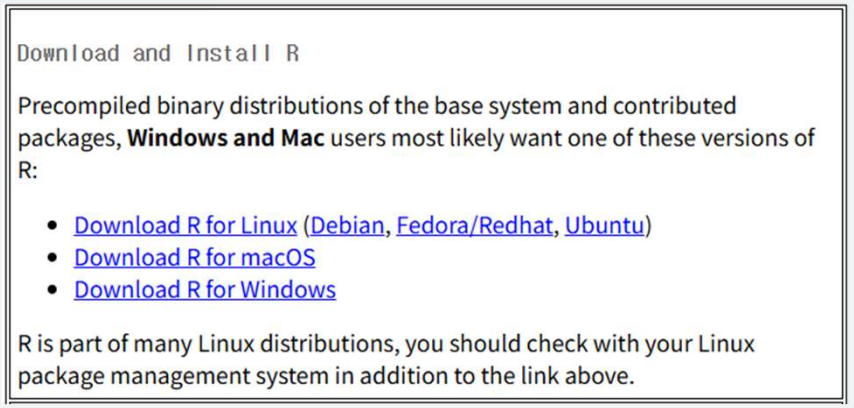
Run the Windows installer (`.exe`)
Keep the default installation options

3 CHECK

Open R and run: **`R.version.string`**
Confirm that R starts without errors

For this workshop: no PATH setup or Rtools installation is needed.

IMAGE PLACEHOLDER


Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- Download R for Linux (Debian, Fedora/Redhat, Ubuntu)
- Download R for macOS
- Download R for Windows

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

https://ftp.yz.yamagata-u.ac.jp/pub/cran/

`> R.version.string` → Confirm the installed R version

```
> R.version.string
[1] "R version 4.6.1 (2026-06-24 ucrt)"
```

KEY POINT · Install R first; RStudio will use the existing R installation.

03 PRACTICE

Installing RStudio Desktop

How do we get the interface for working with R?

1 DOWNLOAD

Go to the RStudio download page
Choose ***"Download RStudio"*** >> click the Download link

2 INSTALL

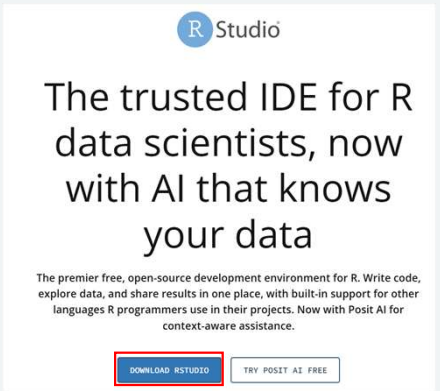
Run the Windows installer (.exe)
Keep the default installation options

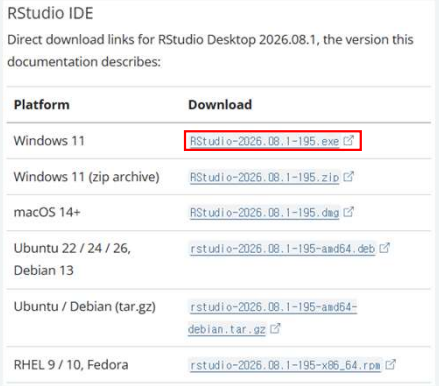
3 LAUNCH

Open RStudio
Check that the Console starts with your installed R version

Important: R must be installed before RStudio. RStudio usually detects the installed R automatically.

IMAGE PLACEHOLDER





RStudio IDE | The Premier Free IDE for R

Troubleshooting

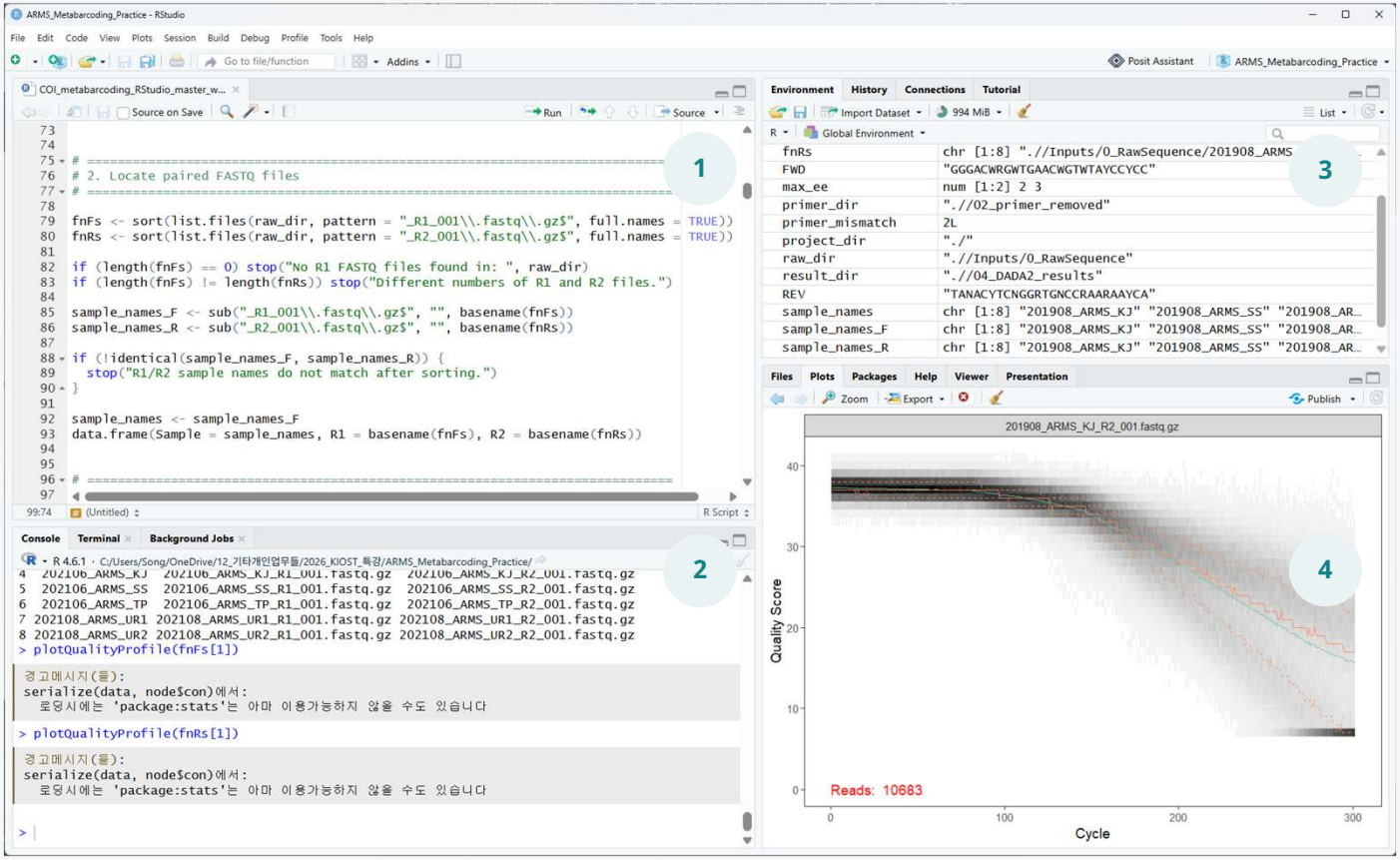
R should be installed before RStudio.

KEY POINT · RStudio is the workspace; it still requires R to perform the analysis.

03 PRACTICE

Meet RStudio

Where do we write code, see results, and manage files?



1 Source
Write and save R scripts

2 Console
Run commands and view messages

3 Environment
Inspect objects created during analysis

4 Files / Plots
Manage files, plots, packages, and help

WORKSHOP TIP **Run selected code with Ctrl + Enter**

KEY POINT · Write code in the Source pane, run it in the Console, and inspect results in the other panes.

RUN A QUICK CHECK

Load `dada2` *and check that the* `data/` *folder is visible.*

ARMS_Metabarcoding_Practice
유형: RStudio Project File

Download: ARMS_Metabarcoding_Practice

1 OPEN THE PROJECT

Open "`ARMS_Metabarcoding_Practice.Rproj`" before running the scripts

2 INSTALL THE PACKAGES

- scripts > open the **"00_install_packages.R"**
- **Run all script code >> automatically installed**

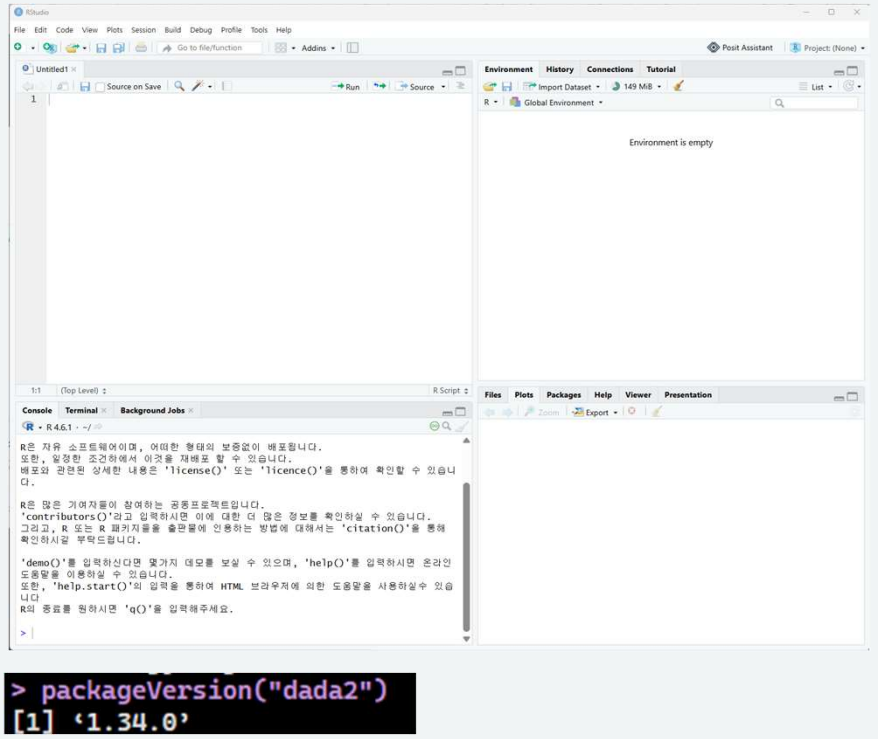
3 RUN A QUICK CHECK

Load dada2 and confirm that the workshop data folders are visible

PACKAGE SETUP

```
install.packages(c("tidyverse", "vegan", "indicspecies"))
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
BiocManager::install("dada2")
library(tidyverse)
library(vegan)
library(indicspecies)
library(dada2)

packageVersion("dada2")
```

IMAGE PLACEHOLDER



KEY POINT · One RStudio Project + the same package set keeps the practical reproducible.

03 PRACTICE

Why do raw reads need processing?

Why problems must we solve before inferring ASVs?

RAW SEQUENCING READS

Biological signal + technical noise

Raw reads are observations — not yet biological sequence variants.

Repeated identical reads are useful abundance information; they can be collapsed computationally while preserving counts.

What must we handle?

Low-quality bases

Trim or filter unreliable positions and reads.

Sequencing / PCR errors

Model technical errors so they are not mistaken for real variants.

Primer / adapter sequence

Remove non-biological sequence before downstream inference.

Chimeric sequences

Identify PCR-derived combinations that do not represent true templates.

Repeated identical reads

Dereplicate identical sequences for efficient computation while preserving read counts.

QUALITY CONTROL + ERROR MODELING → **ASVs**

KEY POINT · Raw reads contain both biological information and technical noise, so they must be processed before ASV inference.

03 PRACTICE

How are ASVs different from OTUs?

Two different strategies for defining sequence units

OTU Cluster similar sequences

A predefined similarity threshold determines which sequences are grouped together.

Sequence A ATGCCGTAACTG
Sequence B ATGCCGTAACTA
Sequence C ATGCCGTAACCG

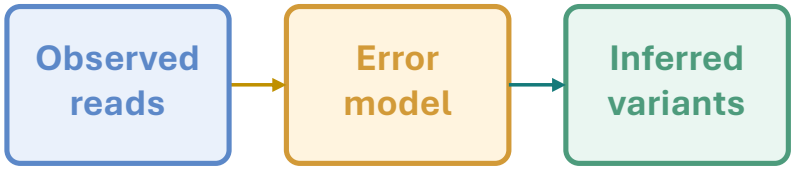
→ **OTU 1**

Example threshold: 97% similarity

The similarity threshold is chosen by the analyst.

ASV Infer sequence variants

DADA2 models sequencing errors and asks whether observed differences are consistent with error or with distinct variants.



ATGCCGTAAC × 124 → ATGCCGTAAC ASV1
ATGTCGTAAC × 3 → ATGTCGTAAC ASV2

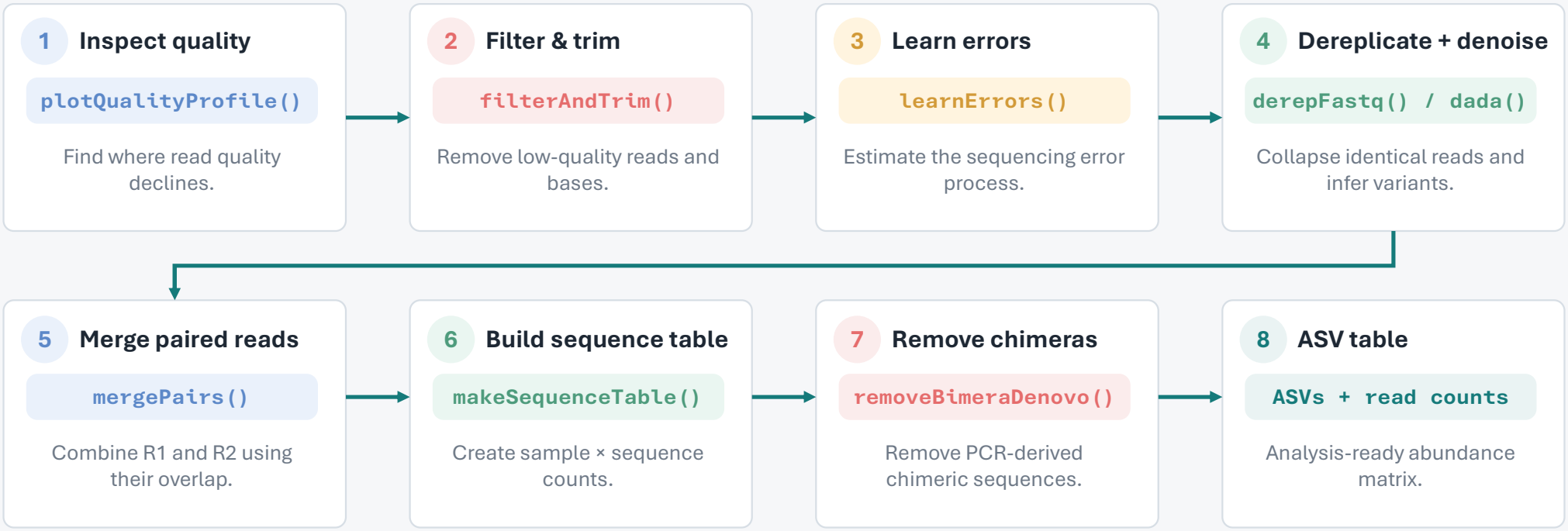
No arbitrary similarity threshold is used to define ASVs.

KEY POINT · OTUs cluster sequences by a chosen similarity threshold; ASVs use error modeling to infer distinct sequence variants.

03 PRACTICE

DADA2 workflow: from FASTQ to ASV table

How do raw sequencing reads become an analysis-ready sequence table?



TODAY'S FOCUS Understand why each step is performed — you do not need to memorize every function or parameter.

KEY POINT · DADA2 progressively reduces technical noise while preserving biological sequence variation.

03 PRACTICE

Practical 1 — Open the workshop project & settings

Let's make sure everyone starts from the same working environment.

1 OPEN THE PROJECT
Double-click `ARMS_Metabarcoding_Practice.Rproj`

2 CHECK THE FOLDERS
Confirm `data/` `scripts/` `intermediate/` `outputs/`

3 OPEN THE SCRIPT
Open `scripts/Day01_DADA2_ASV_inference.R`

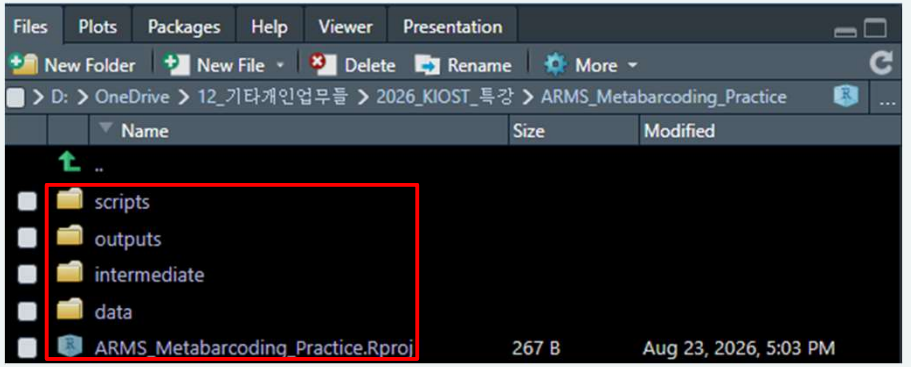
4 RUN THE FIRST CHECK
Run selected code with `Ctrl + Enter`

QUICK CHECK

Expected: R1/R2 FASTQ files

```
library(dada2)
list.files("data/fastq")
```

IMAGE PLACEHOLDER



When the setup is correct

```
> list.files("data/fastq")
 [1] "Bomok1_R1_001.fastq.gz"     "Bomok1_R2_001.fastq.gz"
 [3] "Bomok2_R1_001.fastq.gz"     "Bomok2_R2_001.fastq.gz"
 [5] "Bomok3_R1_001.fastq.gz"     "Bomok3_R2_001.fastq.gz"
 [7] "Bomok4_R1_001.fastq.gz"     "Bomok4_R2_001.fastq.gz"
 [9] "Gangjeong1_R1_001.fastq.gz" "Gangjeong1_R2_001.fastq.gz"
[11] "Gangjeong2_R1_001.fastq.gz" "Gangjeong2_R2_001.fastq.gz"
[13] "Gangjeong3_R1_001.fastq.gz" "Gangjeong3_R2_001.fastq.gz"
[15] "Gangjeong4_R1_001.fastq.gz" "Gangjeong4_R2_001.fastq.gz"
```

KEY POINT · Open the RStudio Project first and confirm the input files before starting the analysis.

Practical 2 — Inspect read quality

Where should we trim without losing paired-end overlap?

QUALITY | FILTER | ERROR | DENOISE | MERGE | CHIMERA | ASV TABLE

What are we looking for?

- Compare quality patterns between R1 and R2
- Find where quality declines near the read ends
- Choose trimming positions for the next step

R CODE

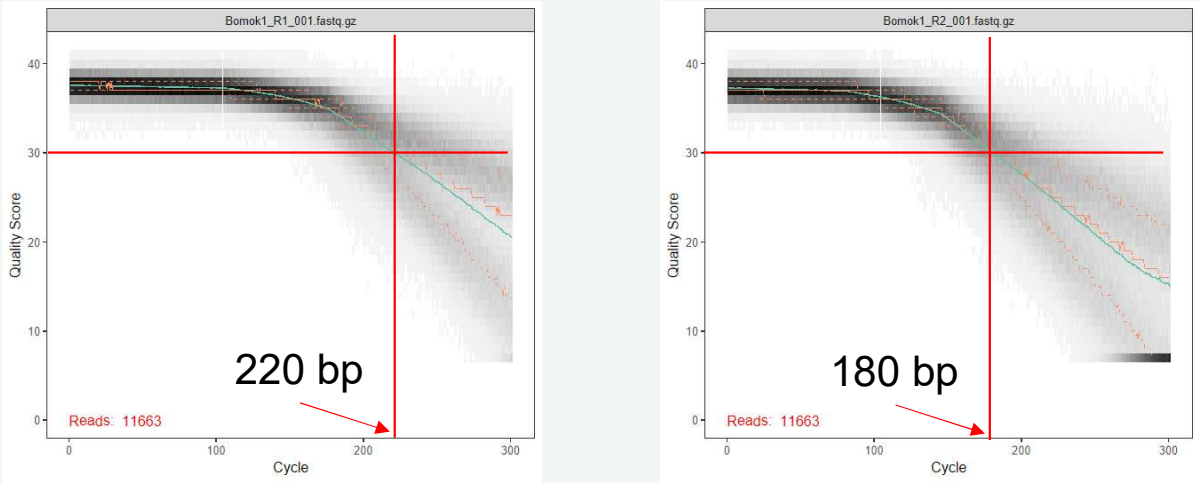
```
# 2. Inspect Read Quality (Raw) -------------
plotQualityProfile(fwd[1:2])
plotQualityProfile(rev[1:2])
```

Do not trim by quality alone

Forward and reverse reads **must still overlap** after trimming.

Keep enough high-quality sequence for **mergePairs()**.

QUALITY PROFILE PLACEHOLDER



ASK THE CLASS

1. Which read loses quality earlier?
2. Where would you truncate R1 and R2?
3. Will enough overlap remain?

KEY POINT · Trimming must balance read quality and paired-end overlap.

03 PRACTICE

Practical 3-5 — Filter and learn errors

How do we remove low-quality reads and model sequencing errors?

QUALITY | **FILTER** | **ERROR** | DENOISE | MERGE | CHIMERA | ASV TABLE

filterAndTrim()

Remove low-quality reads/bases using the trimming and filtering settings selected from the quality profiles.

learnErrors()

Learn the sequencing error pattern directly from the filtered data.

WORKSHOP CODE

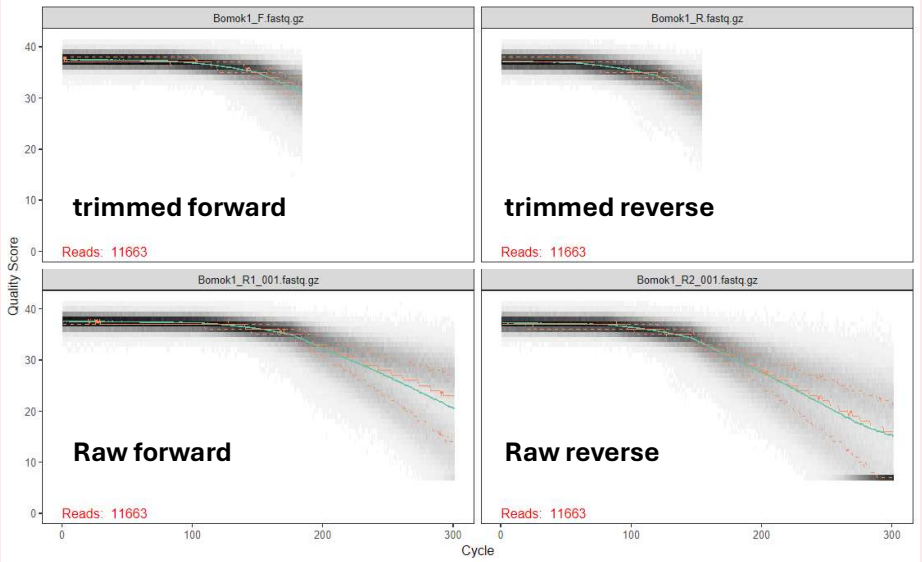
```
# 3. Filter and Trim ----------------------------------------
primer_len <- c(26, 26)  # COI FWD and REV primer lengths
trunc_len <- c(210, 180)

filt_f <- file.path(filt_path, paste0(sample_id, "_F.fastq.gz"))
filt_r <- file.path(filt_path, paste0(sample_id, "_R.fastq.gz"))

filtered <- filterAndTrim(
  fwd, filt_f, rev, filt_r, truncLen = trunc_len, trimLeft = primer_len,
  maxN = 0, maxEE = c(2, 4), truncQ = 2, rm.phix = TRUE, matchIDs = TRUE,
  compress = TRUE, multithread = FALSE)

# 4. Compare Before & After Trimming ------------------------
plotQualityProfile(c(fwd[1], filt_f[1], rev[1], filt_r[1]))

# 5. Learn error rates --------------------------------------
err_f <- learnErrors(filt_f, multithread = threads)
err_r <- learnErrors(filt_r, multithread = threads)

plotErrors(err_f, nominalQ = TRUE); plotErrors(err_r, nominalQ = TRUE)
```

KEY POINT · Filtering removes unreliable reads, and error learning builds the model used for denoising.

Practical 3-5 — Filtering results and learned error model

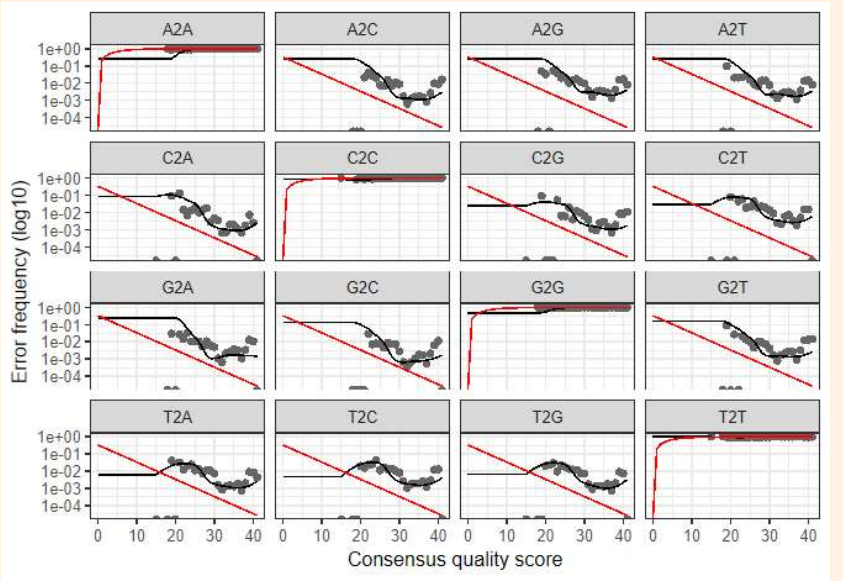
What changed after filtering, and what error pattern did DADA2 learn?

`filterAndTrim()`



Low-quality tails are trimmed, and poor-quality reads are removed before denoising.

`learnErrors()`



DADA2 estimates how error rates change with quality score using the filtered data.

KEY POINT · Filtering removes unreliable reads, and error learning prepares the model used for denoising.

Practical 6 — Dereplication and denoising

How does DADA2 separate sequencing errors from real sequence variants?

QUALITY | FILTER | ERROR | **DENOISE** | MERGE | CHIMERA | ASV TABLE

derepFastq()

Collapse identical reads into unique sequences and record how many times each sequence occurs.

dada()

Use the learned error model to infer sequence variants separately from the forward and reverse reads.

WORKSHOP CODE

```
# 6. Dereplication and ASV inference (Denoise) ------------------------
# 1) Dereplication
derepFs <- derepFastq(filt_f)
derepRs <- derepFastq(filt_r)

# 2) ASV inference
asv_f <- dada(derepFs, err = err_f, multithread = FALSE)
asv_r <- dada(derepRs, err = err_r, multithread = FALSE)
```

KEY POINT · Dereplication summarizes identical reads, and DADA2 uses the learned error model to infer real sequence variants.

Practical 7-9 — Merge reads, remove chimeras, and build the ASV table

How do denoised paired reads become the final ASV table?

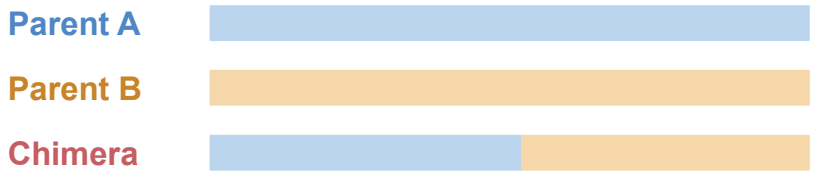
QUALITY | FILTER | ERROR | DENOISE | **MERGE** | **CHIMERA** | **ASV TABLE**

1 · Merge paired reads

`mergePairs()`

Denoised R1 and R2 must overlap and agree in the shared region.

2 · Remove PCR chimeras



`removeBimeraDenovo()`

Removes PCR artifacts formed by combining parts of different parent sequences.

FINAL PROCESSING

```
# 7. Merge paired reads ------------------------------------
min_overlap <- 20

merged <- mergePairs(asv_f, derepFs, asv_r, derepRs,
                     minOverlap = min_overlap, maxMismatch = 0)

names(merged) <- sample_id
sequence_table <- makeSequenceTable(merged)


# 8. Remove chimeras ---------------------------------------
asv_matrix <- removeBimeraDenovo(
  sequence_table, method = "consensus", multithread = threads)

# 9. Export ASV table and representative sequences ----------
abundance_order <- order(colSums(asv_matrix), decreasing = TRUE)
asv_matrix <- asv_matrix[, abundance_order, drop = FALSE]
…
```

KEY POINT · Denoised reads are merged, PCR chimeras are removed, and the remaining sequences form the final ASV table.

03 PRACTICE

What did DADA2 give us?

Output 1 — ASV abundance table

ASV × sample count matrix

Each value is the number of reads assigned to an ASV in a sample.

	Sample 1	Sample 2	Sample 3
ASV1	1250	21	0
ASV2	32	845	102
ASV3	0	15	520
…	…	…	…

ASV table This table is the community matrix used for downstream ecological analysis.

How to read it

ROW — **ASV**
one inferred sequence variant

COLUMN — **Sample**
one biological sample

VALUE — **Read count**
abundance of that ASV in the sample

Day 2 input: community abundance data

KEY POINT · The ASV table tells us how sequence variants are distributed across samples.

03 PRACTICE

What did DADA2 give us?

Output 2 — Representative ASV sequences

Representative sequences (FASTA)

One nucleotide sequence is retained for each ASV.

```
>ASV1
ATGGCACCTTTGACT...

>ASV2
ATAGCTCGTACCAA...

>ASV3
TTGCCGATGGTCAA...
```

Sequence identity only — no organism name yet

Do we know which organism ASV1 is?

Not yet.

DADA2 infers distinct sequence variants,
but it does not assign taxonomy.

Sequence inference
≠ Taxonomic identification

KEY POINT · Representative sequences are the starting point for BLAST and taxonomic assignment on Day 2.

03 PRACTICE

Day 1 — What did we accomplish?

From ARMS samples to an ASV dataset

Today’s workflow

Illumina — Paired-end sequencing

FASTQ — Raw reads + quality

QC + DADA2 — Error-aware sequence inference

ASV outputs — Table + sequences

Take-home messages

1

Raw reads contain technical errors

Quality control is required before treating sequence differences as biological variation.

2

DADA2 uses an error model

It learns sequencing-error patterns and infers ASVs rather than clustering at an arbitrary similarity threshold.

3

Day 1 ends with two key outputs

The ASV abundance table and representative ASV sequences become the starting point for taxonomy and ecology.

WORKSHOP OVERVIEW

Overview — workshop workflow

From a marine sample to ecological interpretation

01 Sampling & DNA Preparation

Collect and prepare the biological material

1. Marine sample collection
2. DNA extraction
3. COI barcode amplification

02 NGS Sequencing

Convert amplified DNA into sequencing reads

1. Library preparation
2. Illumina sequencing
3. Raw sequencing output

03 Bioinformatics Processing

Turn raw reads into reliable sequence variants

1. FASTQ quality control
2. Filtering & denoising
3. ASV inference with DADA2

04 Ecological Analysis

Connect sequence variants to community ecology

1. Taxonomic assignment
2. Alpha & beta diversity
3. Ecological interpretation

DAY 1 · Raw reads → ASVs

DAY 2 · ASVs → Ecology

Day 1 answered “What sequence variants are present?” — **Day 2 asks “What organisms are they, and how do communities differ?”**